

Employee Attendance Complete Flow Guide

Admin setup se Mobile Web aur Biometric attendance tak

यह guide Tenant Admin, UI developer और backend reviewer को एक ही flow में समझाती है कि employee create होने के बाद location, work arrangement, attendance policy, work pattern, override और punch आपस में कैसे जुड़ते हैं।

Main conclusion: Mobile और Web attendance biometric device के बिना चल सकती है। Client-site attendance के लिए active location, employee assignment, effective work arrangement और Mobile-enabled Published Attendance Policy आवश्यक हैं।

Document	Value
System	AxionPro Workforce Management
Subject	Employee attendance configuration and runtime flow
Audience	Tenant Admin HR UI Developer API Developer
Version date	24 September 2026
Language	Simple Hinglish with technical field names

1 Sabse pehle simple meaning

पूरे flow को school example की तरह समझें। हर table या screen की एक अलग responsibility है। एक screen दूसरी screen का duplicate नहीं है।

Part	Simple meaning	Actual responsibility
Employee	Kaun kaam karega	Tenant ka active employee record
Work Location	Kahan attendance lag sakti hai	Employee ko allowed physical locations
Work Arrangement	Normal duty setup kya hai	Work mode primary location aur policy version
Work Pattern	Kaunse din kaise kaam hoga	Weekly schedule office WFH aur weekly off
Mode Override	Special date ka exception	Temporary approved mode/location change
Attendance Policy	Rules kya hain	Mobile Web biometric GPS geofence scope
Attendance Punch	Actual entry	Check In ya Check Out immutable event

Overall order

Tenant locations and Published Attendance Policy



Employee creation



Employee Work Location Assignment



Employee Work Pattern



Employee Work Arrangement



Optional approved Work Mode Override



Employee Check In and Check Out



EmployeeAttendancePunch ledger

2 Tenant ka one time setup

2.1 Locations create karna

Tenant Admin पहले वे सभी places बनाएगा जहाँ employees काम या attendance कर सकते हैं। Head Office, Branch और Client Site सभी TenantLocation records हैं।

Field	Example	Why required
Location name	PANIPAT REFIN	UI और reports में पहचान
Location type	Client Site	Work mode compatibility
Time zone	India Standard Time	सही local work date
Latitude Longitude	Client site coordinates	GPS distance calculation
Geofence radius	200 metres	Allowed boundary
Attendance allowed	Yes	Punch permission
Active and not deleted	Yes	Runtime eligibility

2.2 Attendance Policy create aur publish karna

Policy category ATTENDANCE होगी। Tenant Policy Type और Policy बनाएगा, version configure करेगा और उसे Published करेगा। Work Arrangement हमेशा exact PolicyVersionId से जुड़ेगा।

Configuration	Client site mobile recommendation
Attendance location scope	Assigned Locations
Allow Mobile	Yes
Allow Web	Tenant decision
Allow Biometric	No if device not purchased
Allow Manual	Only when admin correction flow is approved
Allow Work From Home	According to company rule
Office geofence required	Yes

Configuration	Client site mobile recommendation
Remote GPS required	According to company rule
Outside location with approval	According to approval process

नई client site के लिए हमेशा नई policy बनाना जरूरी नहीं है। Existing Published policy में Mobile, Assigned Locations और required geofence rules सही हैं तो वही version use हो सकता है। Rules बदलें तो existing policy का नया version publish करें।

3 Employee add hone ke baad admin ka exact flow

Step 1 Employee create karein

Employees screen से employee बनता है। Attendance API request में employee ID नहीं लेती; login token से authenticated employee पहचानती है।

Required state	Expected value
Tenant ownership	Same authenticated Tenant
Employee status	Active
Soft deleted	False
Login	Employee self service के लिए active credentials

Step 2 Employee Work Location Assignment

यह screen बताती है employee किन locations पर attendance कर सकता है। एक employee की multiple allowed locations हो सकती हैं, लेकिन effective date windows और primary location rules valid होने चाहिए।

Example assignment	Primary	Attendance	Effective period
Jabalpur Head Office	Yes	Allowed	12 Sep 2026 onward
PANIPAT REFIN Client Site	No	Allowed	30 Sep 2026 onward

Assignment validation

- Employee और location same Tenant के हों।
- Location active और non deleted हो।
- IsAttendanceAllowed true हो।
- Effective From और Effective To valid हों।
- Same employee location की overlapping active assignment duplicate न हो।
- Conflicting primary location windows न हों।

Step 3 Employee Work Pattern

Pattern normal weekly calendar है। इससे expected Office day, WFH day और weekly off पता चलता है।

Day	Example plan
-----	--------------

Day	Example plan
Monday	Client Site
Tuesday	Client Site
Wednesday	Client Site
Thursday	Client Site
Friday	Client Site
Saturday	Weekly Off
Sunday	Weekly Off

Current boundary: Pattern CRUD/configuration मौजूद है, लेकिन वर्तमान mark attendance engine अभी weekly off, planned Office/WFH day, reporting time या late status enforce नहीं करता।

Step 4 Employee Work Arrangement

Arrangement employee का long running normal duty setup है। यह Work Mode, Primary Location और Published Attendance Policy Version को जोड़ता है।

Field	Client deployment example
Employee	Rahul Sharma
Work mode	Client Site
Primary location	PANIPAT REFIN
Attendance policy	Published Mobile Client Site version
Effective From	01 Oct 2026
Effective To	31 Dec 2026
Active	Yes

Arrangement validation

- Client Site mode के साथ Client Site location ही compatible हो।
- Primary location employee की active attendance enabled assignment हो।
- Location assignment पूरे arrangement period को cover करे।

- Policy version Published और effective हो।
- Policy physical location और selected source allow करे।
- एक दिन पर exactly one active effective arrangement हो।

4 Work Mode Override kab use hoga

Override temporary exception है। यह permanent Work Arrangement को edit नहीं करता। उदाहरण: employee normally Client Site पर है लेकिन एक दिन WFH approved है।

Normal setup	Temporary event	Override
Client Site	Site closed on 15 Oct	15 Oct Work From Home
Work From Home	Meeting at Head Office	20 Oct Office at Head Office
Head Office	Short client visit	Specified date Client Site

Expected runtime priority

Approved active override for the date



Effective Work Pattern day



Effective Work Arrangement



Published Attendance Policy configuration

Current boundary: mark attendance repository अभी Work Mode Override को read नहीं करती। इसलिए saved override अभी punch validation में base arrangement को replace नहीं करता। यह runtime integration अलग से complete करना होगा।

5 Biometric device nahi liya to kya hoga

Mobile और Web attendance के लिए biometric device खरीदना या Employee Device Enrollment करना जरूरी नहीं है। Attendance Policy में Mobile या Web allowed होना चाहिए।

Situation	Result
No biometric device but Mobile allowed	Employee mobile से punch कर सकता है
No biometric device but Web allowed	Employee web से punch कर सकता है
Policy only allows Biometric	Mobile Web punch reject होगा
Biometric configured and employee enrolled	Trusted biometric ingestion flow use होगा
Biometric configured but employee not enrolled	उस device से employee match नहीं होगा

Attendance device type master

Code	Meaning	Physical registration
MOBILE	Authenticated mobile app	No
WEB	Authenticated web client	No
BIOMETRIC	Registered physical device	Yes
MANUAL	Authorized admin entry	No

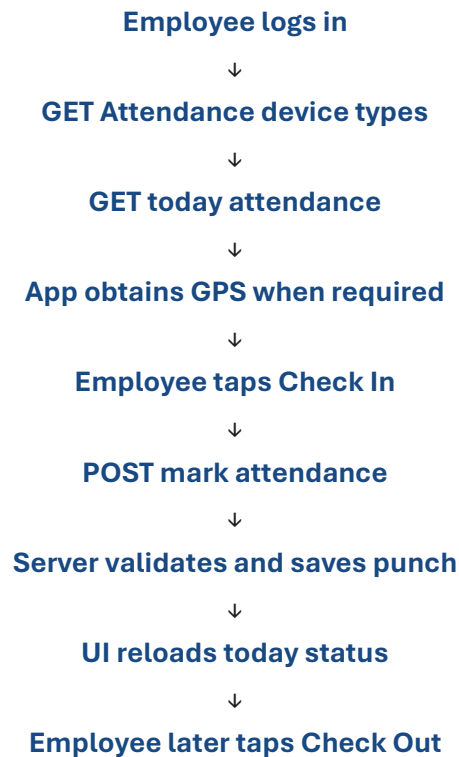
UI GET /api/Attendance/device-types से IDs लेगी / Numeric IDs hard code नहीं किए जाएंगे /

6 Client site par Mobile attendance ka complete example

Configuration checklist

- PANIPAT REFIN को Client Site TenantLocation बनाएं।
- Location पर attendance allowed, timezone, latitude, longitude और geofence radius configure करें।
- Published Attendance Policy में Mobile और Assigned Locations allow करें।
- Employee को PANIPAT REFIN location assign करें।
- Employee को Client Site Work Arrangement बनाकर PANIPAT REFIN primary रखें।
- Effective dates assignment, arrangement और policy में compatible रखें।
- Work Pattern में applicable working days रखें।

Employee app flow



Sample request

POST /api/Attendance/mark-attendance

```

{
  "action": 1,
  "attendanceDeviceTypeId": 1,
  "tenantLocationId": 25,
  "latitude": 29.3909,
  "longitude": 76.9635,
  "accuracyMeters": 15,
  "clientOccurredAt": "2026-10-01T09:05:00+05:30",
  "idempotencyKey": "661a418b-4294-4f87-bd82-d3f08712ba27"
}
  
```

action 1 = Check In और action 2 = Check Out / IDs केवल examples हैं; UI lookup APIs से actual IDs लेगी।

7 Punch ke waqt backend ki validation

No	Validation	Pass condition
1	Bearer token	Request authenticated हो
2	Employee identity	Token का active non deleted employee मिले
3	Idempotency	Same button retry duplicate row न बनाए
4	Arrangement	आज exactly one effective arrangement हो
5	Policy configuration	Arrangement policy version की typed attendance config मिले
6	Device type	Active MOBILE या WEB source हो
7	Policy source	Selected source policy में allowed हो
8	Location	Physical mode में active attendance enabled location हो
9	Location scope	Primary या assigned location rule pass हो
10	Effective dates	Assignment आज valid हो
11	GPS	Required होने पर latitude longitude मिलें
12	Geofence	Employee permitted radius में हो
13	Work date	Location timezone से local date निकले
14	Punch order	Duplicate Check In और orphan Check Out reject हों
15	Persistence	Immutable punch transaction में save हो

8 Data kahan save hota hai

AttendancePolicyVersionConfiguration

यह rules table है। इसमें employee punches नहीं आते। हर Published Attendance Policy version के Mobile, Web, Biometric, Manual, WFH, GPS, geofence और location scope rules रहते हैं।

EmployeeAttendancePunch

हर successful Check In या Check Out की immutable entry यहाँ save होती है।

Column	Purpose
TenantId	Tenant isolation
EmployeeId	Authenticated employee
EmployeeWorkArrangementId	Applied arrangement
PolicyVersionId	Applied policy version
TenantLocationId	Physical location when applicable
AttendanceDeviceTypeId	Mobile Web Biometric or Manual source
WorkDate	Location timezone local date
OccurredAtUtc	Authoritative server time
PunchAction	Check In or Check Out
Latitude Longitude	Captured GPS
AccuracyMeters	Device reported accuracy
DistanceFromLocationMeters	Calculated geofence distance
IdempotencyKey	Safe retry key

Today status endpoint

GET /api/Attendance/today आज की ordered punch timeline और isCurrentlyCheckedIn देता है। UI page load और successful punch के बाद इसे call करेगी।

9 Common scenarios aur expected result

Scenario	Expected result
Mobile allowed no biometric purchased	Mobile punch accepted after other validations
Mobile disabled in policy	Rejected
Client location not assigned	Rejected for Assigned Locations scope
Location attendance disabled	Rejected
Employee outside geofence	Rejected
GPS missing when required	Rejected
Second Check In before Check Out	Rejected
Check Out without Check In	Rejected
Network retries same idempotency key	Original result reused no duplicate
Inactive employee	Rejected
No effective arrangement	Rejected
Two overlapping effective arrangements	Conflict rejected

10 Admin operational checklist

- Employee active और login enabled है।
- Required Tenant locations active हैं।
- Locations की timezone और geofence complete है।
- Attendance Policy version Published है।
- Required Mobile Web या Biometric source allowed है।
- Employee location assignment effective है।
- Primary assignment arrangement period cover करती है।
- Work Arrangement में correct mode location और PolicyVersion selected है।
- Work Pattern configured है।
- Temporary changes approved Override में हैं।
- UI device type IDs GET API से ले रही है।
- Check In और Check Out explicit action भेजे जा रहे हैं।

11 Current completion status and remaining runtime integration

Area	Status	Meaning
Location and arrangement consistency	Implemented	Cross record and effective date validation
Typed attendance policy config	Implemented	Version owned channel location and GPS rules
Mobile Web punch ledger	Implemented	Token identity geofence idempotency and punch order
Device type lookup and FK	Implemented	Seeded catalogue and GET API
Today status	Implemented	Current state and timeline
Work Pattern enforcement during punch	Pending integration	Weekly off planned mode timing and late rules
Override resolution during punch	Pending integration	Approved temporary mode/location must take priority
Biometric log ingestion	Separate flow	Requires trusted configured device and employee enrollment
Manual correction workflow	Separate flow	Needs authorized admin audit and approval rules

यह distinction जरूरी है: configuration screens मौजूद होना और mark attendance runtime में उनका enforce होना अलग बातें हैं। UI को केवल implemented runtime behavior पर भरोसा करना चाहिए।

12 Final simple example

Rahul को Panipat client site भेजना है और biometric device नहीं खरीदा गया है। Admin Panipat को attendance enabled Client Site बनाता है, Mobile enabled Published Attendance Policy चुनता है, Rahul को Panipat assign करता है और Client Site arrangement बनाता है। Rahul mobile app में login करके GPS के साथ Check In करता है। API token, employee, arrangement, policy, source, assigned location और geofence check करके EmployeeAttendancePunch में entry बनाती है। शाम को explicit Check Out दूसरी entry बनाता है। Biometric enrollment इस flow में कहीं आवश्यक नहीं है।